

BRISTOL FILTON, United Kingdom

WMO: 036280

Lat: 51.517N

Lon: 2.583W

Elev: 59

StdP: 100.62

Time Zone: 0.00 (EUW)

Period: 01-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-3.1	-1.7	-5.7	2.3	-0.8	-4.5	2.6	0.1	23.1	10.7	20.3	9.2	2.4	30	0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	26.1	18.6	24.0	17.6	22.3	16.9	19.6	24.0	18.5	22.6	17.6	21.1	4.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.1	21.1	17.0	12.2	19.9	16.2	11.6	19.0	56.1	24.0	52.5	22.8	49.8	21.1	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
14.3	11.5	9.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-5.6	29.2	1.7	2.6	-6.8	31.1	-7.8	32.6	-8.8	34.1	-10.0	36.0
			-5.9	21.8	1.6	2.4	-7.1	23.5	-8.1	25.0	-9.0	26.4	-10.2	28.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	5.4	5.4	7.1	9.8	12.6	15.7	17.4	16.9	14.9	11.9	8.1	5.7
	DBStd	5.20	3.30	3.07	2.89	2.66	2.61	2.32	2.46	2.03	2.22	2.73	3.15	3.59
	HDD10.0	628	143	129	96	34	5	0	0	0	13	71	137	
	HDD18.3	2735	400	361	348	256	179	84	48	55	105	200	307	391
	CDD10.0	973	2	2	6	28	84	172	231	213	147	71	14	4
	CDD18.3	38	0	0	0	1	6	20	9	2	0	0	0	0
	CDH23.3	247	0	0	0	1	10	48	137	42	9	2	0	0
	CDH26.7	41	0	0	0	0	0	8	26	7	0	0	0	0
(14) Wind	WSAvg	4.5	5.1	5.1	4.8	4.6	4.8	4.4	4.2	3.9	3.9	4.4	4.3	4.6
(15) Precipitation	PrecAvg	925	99	68	56	62	68	61	64	77	68	101	101	94
	PrecMax	1258	193	161	107	180	139	167	154	148	141	195	214	174
	PrecMin	656	4	3	14	11	11	10	9	6	15	44	38	23
	PrecStd	149	45	43	27	39	32	39	34	44	37	42	41	45
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	13.2	17.4	21.9	24.7	27.7	29.3	27.3	24.7	20.0	15.4	14.1
		MCWB	11.6	11.5	11.1	14.0	17.2	18.6	19.3	21.5	18.7	16.6	13.9	12.5
	2%	DB	11.9	11.7	14.6	18.6	21.9	24.3	26.8	24.0	22.0	17.7	14.3	13.0
		MCWB	10.9	10.0	10.2	12.9	15.9	17.4	18.6	18.8	16.8	14.9	13.0	11.9
	5%	DB	11.0	10.7	12.8	16.2	19.5	22.5	24.5	22.1	20.3	16.7	13.5	11.9
		MCWB	9.9	9.1	9.9	11.4	14.7	16.8	17.8	17.5	15.9	14.4	12.2	10.7
	10%	DB	10.1	9.9	11.5	14.3	17.4	20.4	22.3	20.7	19.0	15.8	12.7	10.9
		MCWB	9.1	8.7	9.1	10.5	13.3	15.7	16.9	16.5	15.4	13.8	11.4	9.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	11.9	12.2	15.0	18.7	20.7	20.9	23.7	19.5	17.1	14.5	12.9
		MCDB	12.5	13.0	14.7	19.8	22.4	25.6	27.2	26.0	23.0	19.1	15.0	13.6
	2%	WB	11.2	10.5	11.2	13.3	16.6	18.5	19.2	19.5	17.9	15.8	13.4	12.1
		MCDB	11.7	11.3	13.4	17.4	20.8	22.7	25.1	22.5	20.6	17.1	14.0	13.0
	5%	WB	10.1	9.7	10.4	12.1	15.1	17.1	18.4	18.2	16.9	15.0	12.5	11.0
		MCDB	10.8	10.4	12.4	15.2	18.4	21.1	23.4	21.0	19.3	16.2	13.3	11.8
	10%	WB	9.3	8.9	9.6	11.1	14.0	16.2	17.4	17.4	16.0	14.2	11.5	10.0
		MCDB	10.1	9.8	11.3	13.9	17.0	19.5	21.3	19.8	18.3	15.6	12.5	10.8
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	5.2	6.8	7.9	7.7	7.7	7.7	7.2	7.5	6.1	5.4	5.0
		MCDBR	5.2	5.1	8.5	10.8	11.0	11.2	11.4	9.3	9.3	6.5	5.4	5.1
	5% WB	MCWBR	5.1	4.2	5.5	6.3	6.2	5.9	5.2	5.2	5.3	4.4	4.6	5.1
		MCWBR	5.1	4.6	7.3	9.3	9.8	9.4	9.7	8.3	8.0	5.7	5.2	5.0
(40) Clear-Sky Solar Irradiance	taub	0.327	0.342	0.375	0.396	0.400	0.393	0.390	0.377	0.367	0.353	0.327	0.316	
		taud	2.355	2.376	2.284	2.242	2.268	2.320	2.333	2.387	2.410	2.422	2.404	2.333
	Ebn at Noon	676	777	813	840	853	862	858	851	818	753	673	620	
		Edh at Noon	64	81	106	124	127	122	118	106	93	77	60	55
(44) All-Sky Solar Radiation	RadAvg	0.78	1.47	2.61	4.01	4.93	5.35	5.01	4.15	3.12	1.77	0.94	0.61	
	RadStd	0.07	0.18	0.33	0.46	0.25	0.57	0.48	0.36	0.23	0.16	0.09	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (77 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page